

FlexRoads – Working with DataFlex Libraries: Secure, Reusable, Powerful

Instructions for the hands-on assignments

Part 1 – Quill Editor Setup

1.1 Download or clone the workspace from Git

1. Go to: <https://github.com/siam-smeets/DataFlexLibraries>
2. Clone the repository or download it as a ZIP.

1.2 Open and test run the application

1. Open the DataFlex Libraries workspace in the DataFlex 2026 studio
2. Register the Web Application
3. Run the Web Application.

1.3 Add the Quill Editor Package

1. Open the Package Manager (Tools -> Package Manager or CTRL + SHIFT + P)
2. Open the "BROWSE" tab
3. Search for "Quill Editor"
4. Click on install.
5. Verify that in the dfpkg folder the Quill Editor is installed and in the AppHtml the RichTextEditor folder exists.

1.4 Update Views

1. Replace cWebEdit to cWebRichTextEditor in FeedbackZoom
2. Replace cWebHtmlBox to cWebRichTextViewer in FeedbackSelect
3. Run and test the application and see if you can add better looking texts in the editor and see them correctly in the select view.

Part 2 – XSS test

2.1 Insert XSS in the Quill Editor

1. Open the FeedbackZoom in the Studio
2. Set pbAllowShowHtml to True
3. Recompile and open the WebApp in the browser
4. Open the HTML via the HTML button in the toolbar of the editor
5. Paste the following in the Editor
`<p onmouseover="alert(1)" style=display:block>test</p>`
6. Give it a subject and save the record.
7. Now open the Feedback Select and hover over the text. As you can see it will immediately execute the alert.

Part 3 – XSS Sanitizer

3.1 Adding XSS Sanitizer to the WebApp

1. Open the WebApp.src
2. Add Use oXssSanitizerStandard.pkg within WebApp object
3. Add the following code below the use statement
Send SetForbiddenAttributes of ghoSanitizer “onclick,onmouseover”
4. Open the cUserFeedbackDataDictionary.dd
5. Implement the Save_Main_File
6. Add the following code before the Forward Send
String sDescription

Move UserFeedback.Description to sDescription
Get Sanitize of ghoSanitizer sDescription to sDescription
Move sDescription to UserFeedback.Description
7. Recompile and open the WebApp in the browser
8. Try step 2 again by creating a new record with the alert.
9. Now open the Feedback Select and notice that the onmouseover is removed from the HTML and the alert is not fired anymore.

Part 4 – Uploading File

4.1 Upload a malicious file

1. Open the WebApp in the browser
2. In the UploadZoom try to upload a malicious file. This can range from a .png that contains text, uploading an EICAR file (test virus file).
3. See that the upload of a malicious file is possible
4. Now add the following use statement to the UploadZoom
Use SecureUpload\cWebSecureFileUploadForm.pkg
5. And change oWebFileUploadForm1 to cWebSecureFileUploadForm
6. Recompile and open the WebApp in the browser
7. Now try uploading the same file again and see that you now will get an error message.

Part 5 – SecureUpload to Quill

5.1 Add SecureUpload to the FeedbackZoom

1. Open the FeedbackZoom and add the following use statement before the cWebRichTextEditor use statement
Use SecureUpload\cWebSecureBaseFileUpload.pkg
2. Recompile and open the WebApp in the browser
3. Try uploading an image that contains text. And see that you are not allowed to upload the image anymore.

Part 6 – Configurable Settings

6.1 Add configurable settings

1. Copy the contents of part 6.zip over your workspace
2. Add the settings table using Database -> Add Existing table
3. Now create a subclass of the cWebSecureFileUploadForm and implement the OnUploadSettings and update the settings based on the Settings table
4. Recompile and open the webapp in the browser
5. Open the SettingsZoom and update one of the settings (e.g. Check Contents)
6. And save it and try to upload a file that was previously not allowed and should now be allowed by the settings change.